Joint Attention-Field × PRF model

anatomy of the parameters

Forward model (per voxel v, per HP condition C):

$$R_{v\_C}(g) = M_C(g) \cdot S_v(g)$$

$$M_C(g) = 1 + \mathrm{sign} \cdot \left(g_{HP} \cdot A_{H_C}(g) + g_{LP} \cdot \Sigma_{\square \neq H_C} A_{\square}(g) \right)$$

$$A_{\square}(g) = \exp\left(-\|g - \mu_{\square}\|^2 / (2\, \sigma_{AF}^2) \right) \qquad [\text{ unit peak }]$$

$$S_v(g) = \text{Gaussian PRF at } (x_v, y_v) \text{ with width } sd_v$$

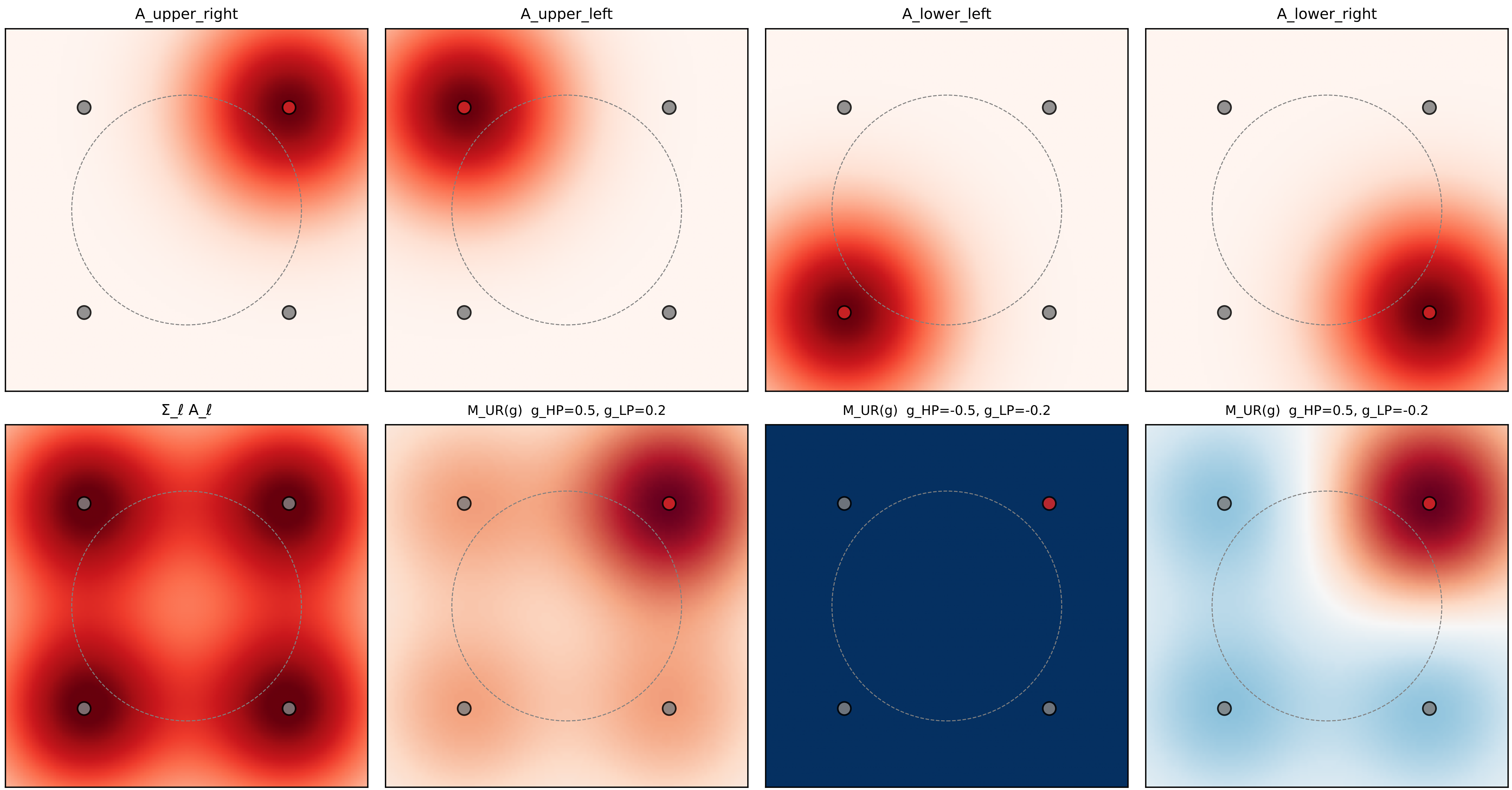
Shared parameters (same value for every voxel in a ROI):

- σ_{AF} — width of each AF Gaussian – how spatially focused the modulation is around each ring location.
- g_{HP} — modulation strength AT THE HP LOCATION (negative = suppression, positive = attraction)
- g_{LP} — modulation strength AT EACH OF THE 3 LP LOCATIONS (same convention as g_{HP}).

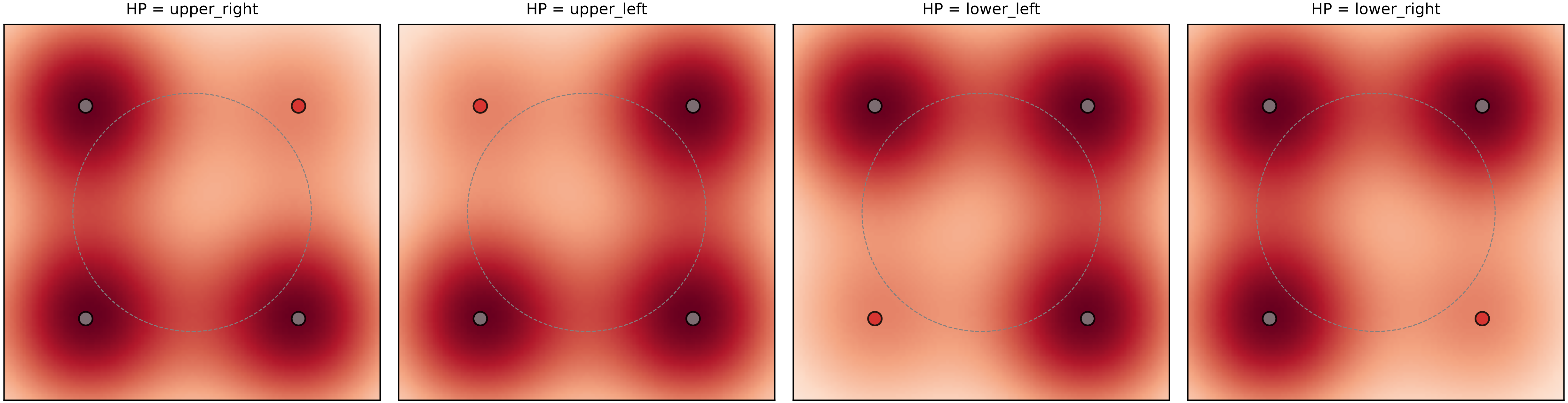
Per-voxel parameters: x, y, sd, baseline, amplitude (Gaussian PRF).

Quantity of interest: $g_{HP} - g_{LP}$ or $\log((1+g_{HP})/(1+g_{LP}))$
negative ⇒ HP-specific suppression (vs surrounding LP).

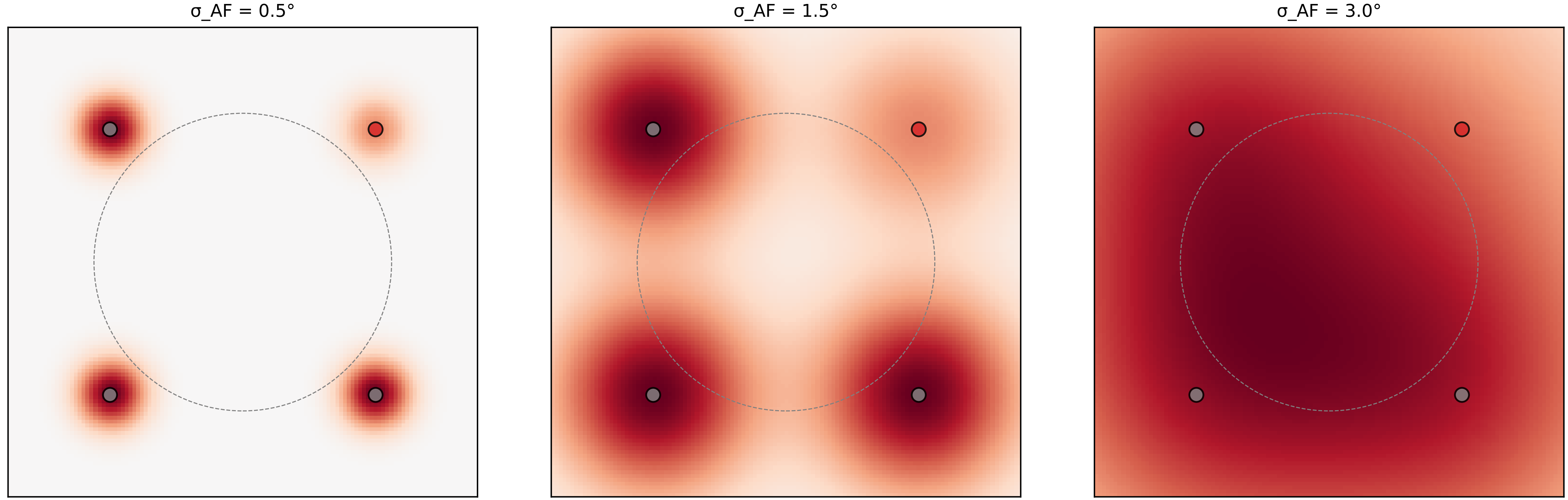
(2/7) The four ring attention fields A_ℓ(g) (σ_AF = 1.92°)



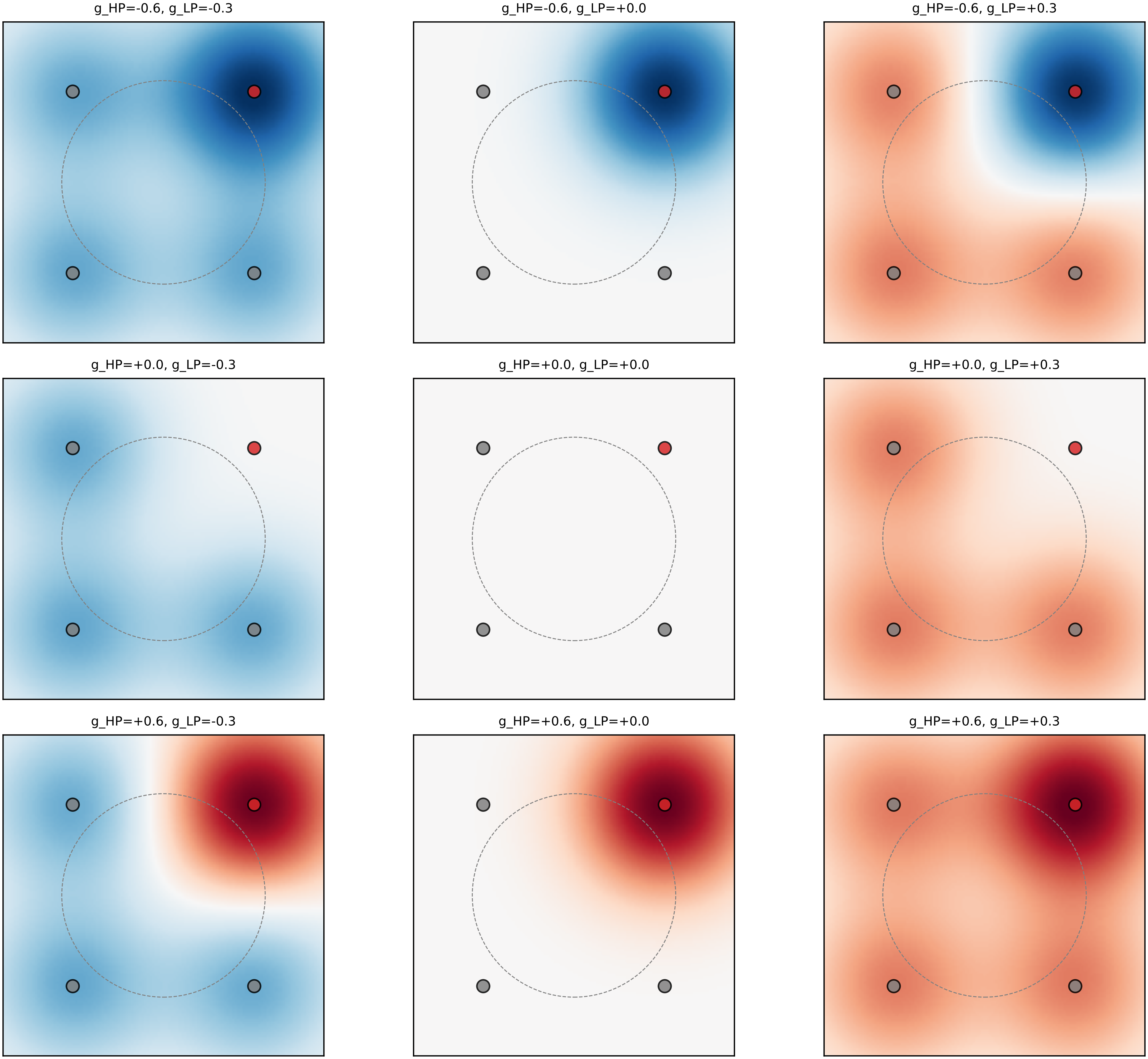
(3/7) M_C(g) for each HP condition (σ_AF=1.92, g_HP=0.14, g_LP=0.29; this example: HP suppressed)



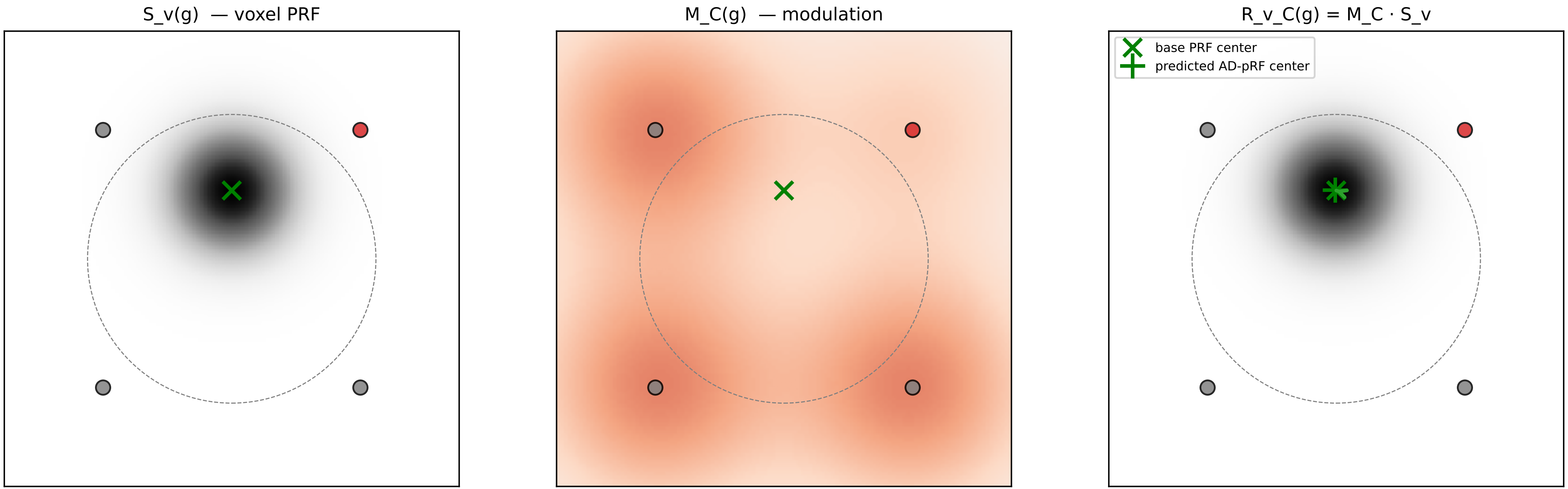
(4/7) σ_AF effect — same gains (g_HP=0.14, g_LP=0.29), HP=UR.
Smaller σ_AF → spatially focused modulation; larger σ_AF → diffuse / overlapping fields.



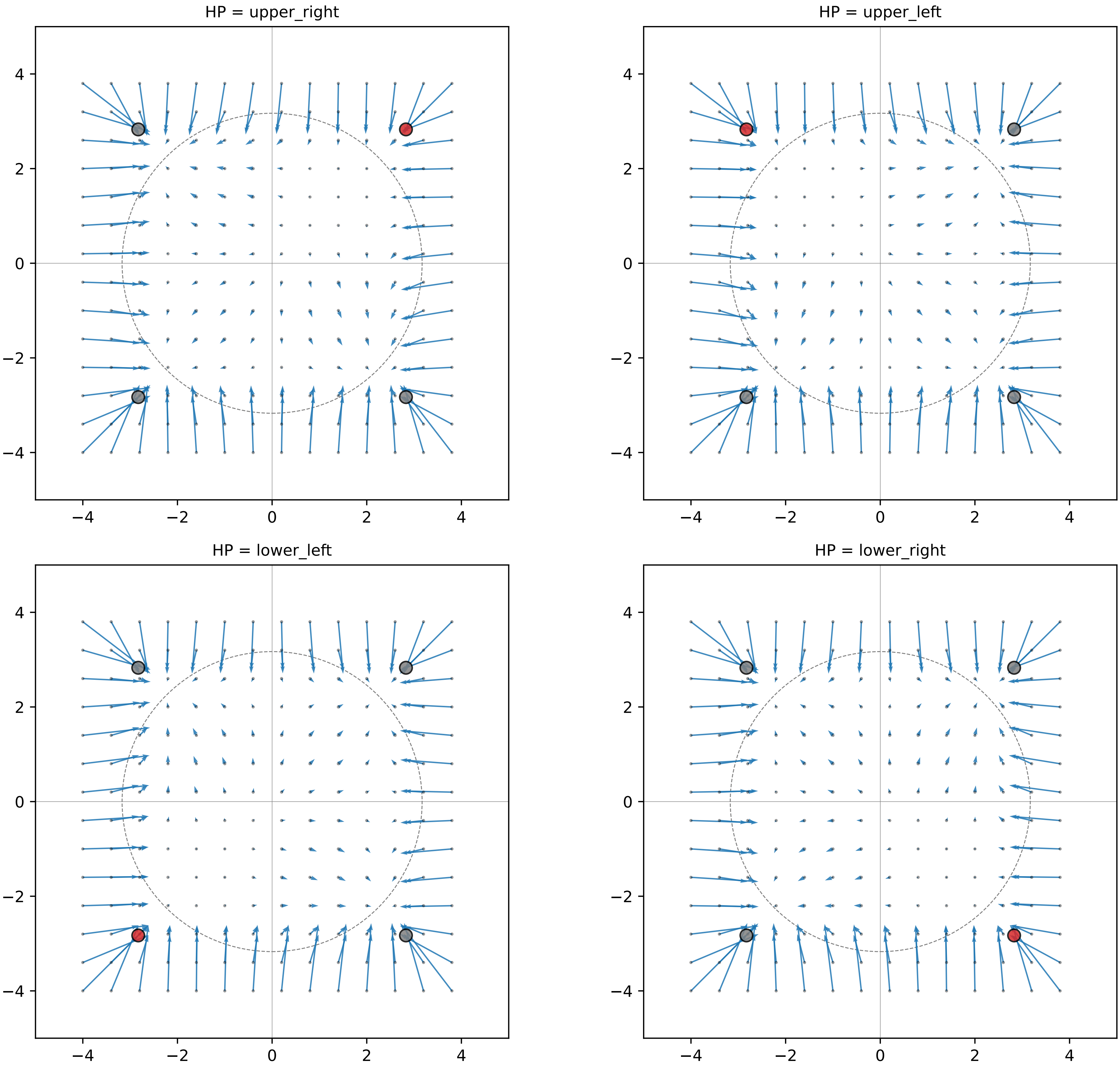
(5/7) Gain sweep — σ_AF = 1.92°, HP = UR.
Rows: g_HP (HP-location strength). Columns: g_LP (LP-location strength).
Negative ⇒ suppression (blue), positive ⇒ attraction (red).



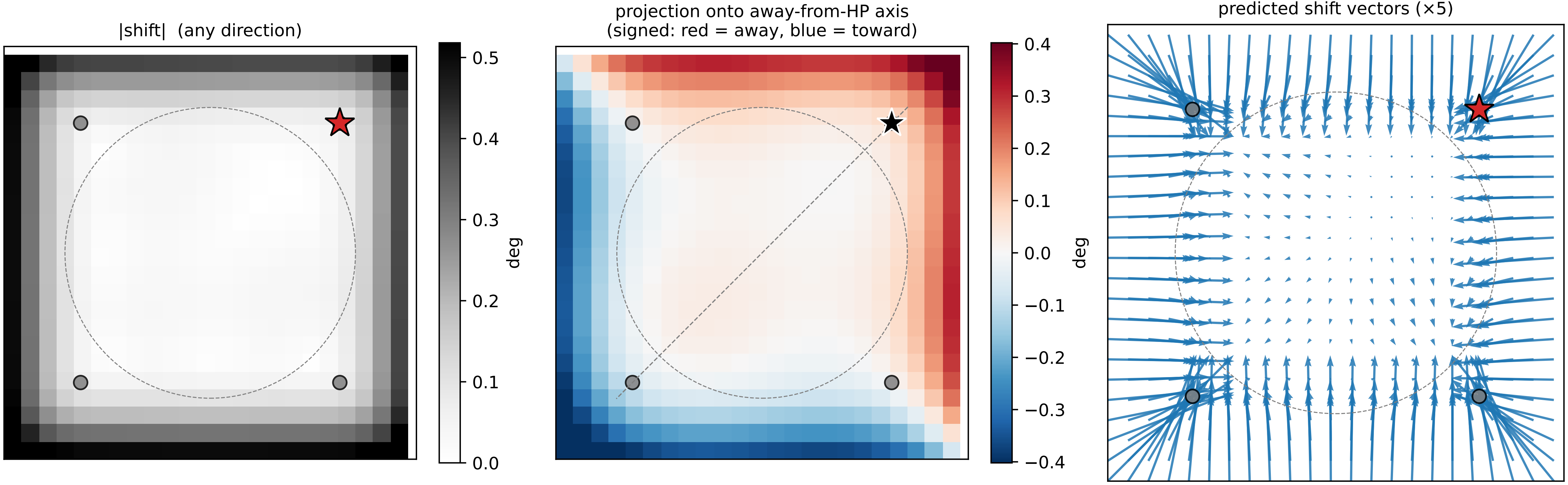
(6/7) Single-voxel example. PRF at (+0.0, +1.5), sd=1.0. HP = upper_right. σ_AF=1.92, g_HP=0.14, g_LP=0.29.

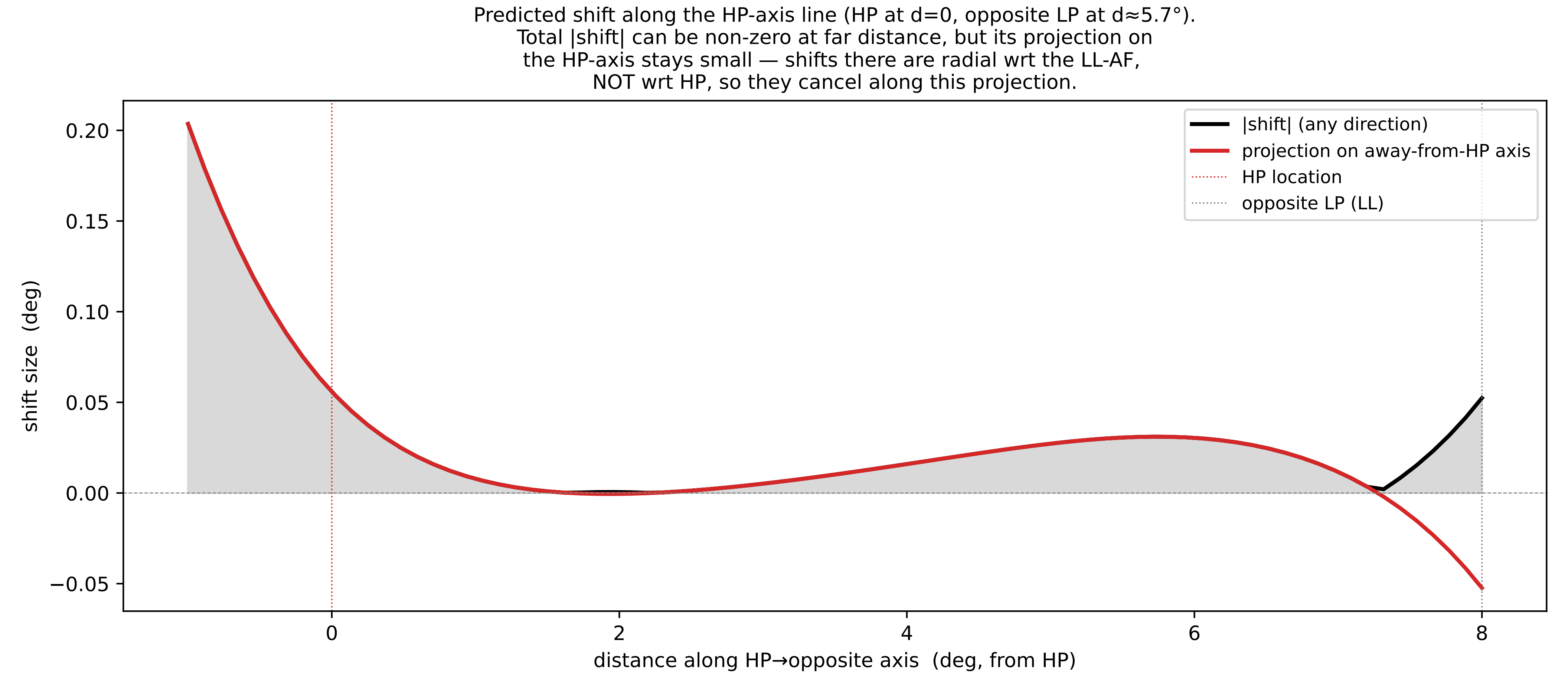


(7/7) Predicted PRF-center shift field (full FOV) (σ_AF=1.92, g_HP=0.14, g_LP=0.29, prf_sd=1.0)
Each arrow goes from a seed PRF center to the predicted AD-pRF center (arrows scaled ×5.0).

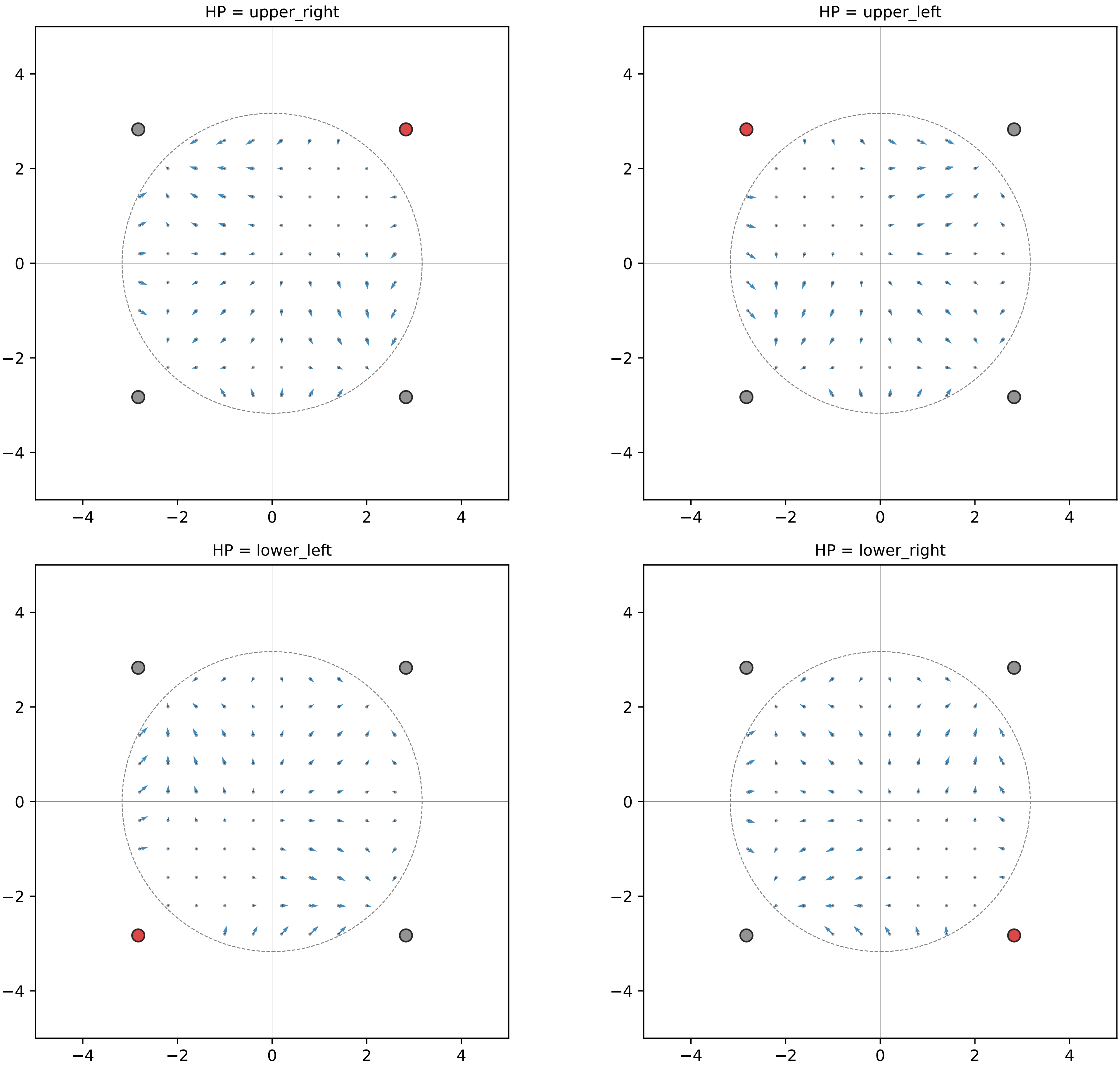


Why projection on the away-from-HP axis is small at large distance.
σ_AF=1.92, g_HP=+0.14, g_LP=+0.29, prf_sd=1.0. HP = upper-right (red star).

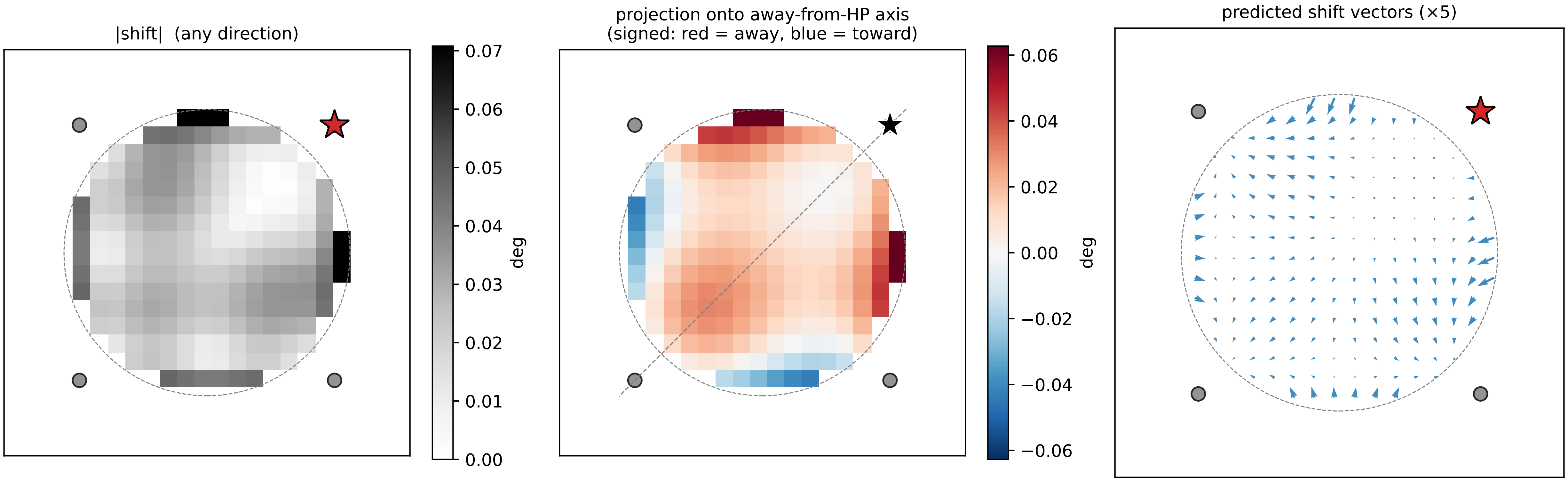


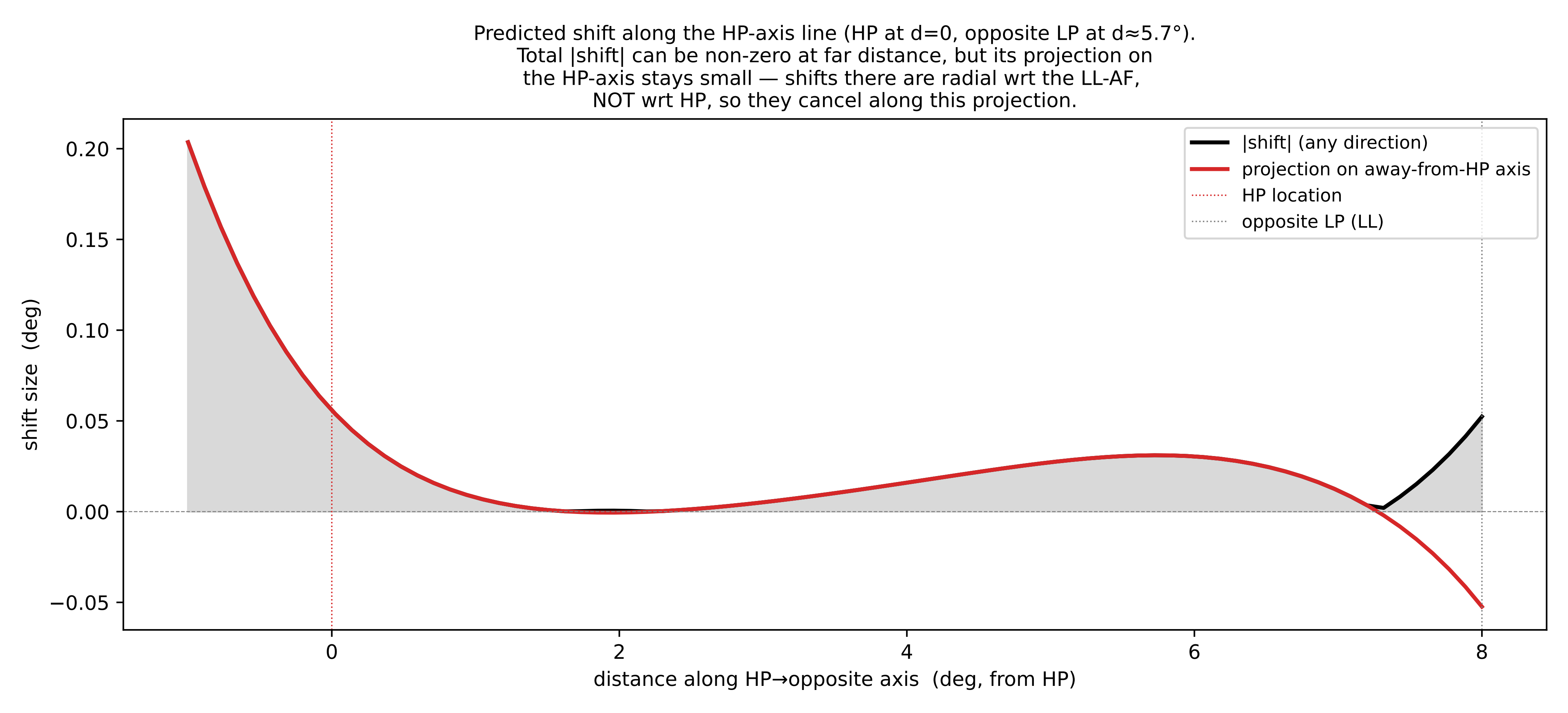

Predicted shift along the HP-axis line (HP at d=0, opposite LP at d≈5.7°).
Total |shift| can be non-zero at far distance, but its projection on
the HP-axis stays small — shifts there are radial wrt the LL-AF,
NOT wrt HP, so they cancel along this projection.
|shift| (any direction)
projection on away-from-HP axis
HP location
opposite LP (LL)
shift size (deg)
0.20
0.15
0.10
0.05
0.00
−0.05
0
2
4
6
8
distance along HP→opposite axis (deg, from HP)

(7/7) Predicted PRF-center shift field (inside aperture only) (σ_AF=1.92, g_HP=0.14, g_LP=0.29, prf_sd=1.0)
Each arrow goes from a seed PRF center to the predicted AD-pRF center (arrows scaled ×5.0).

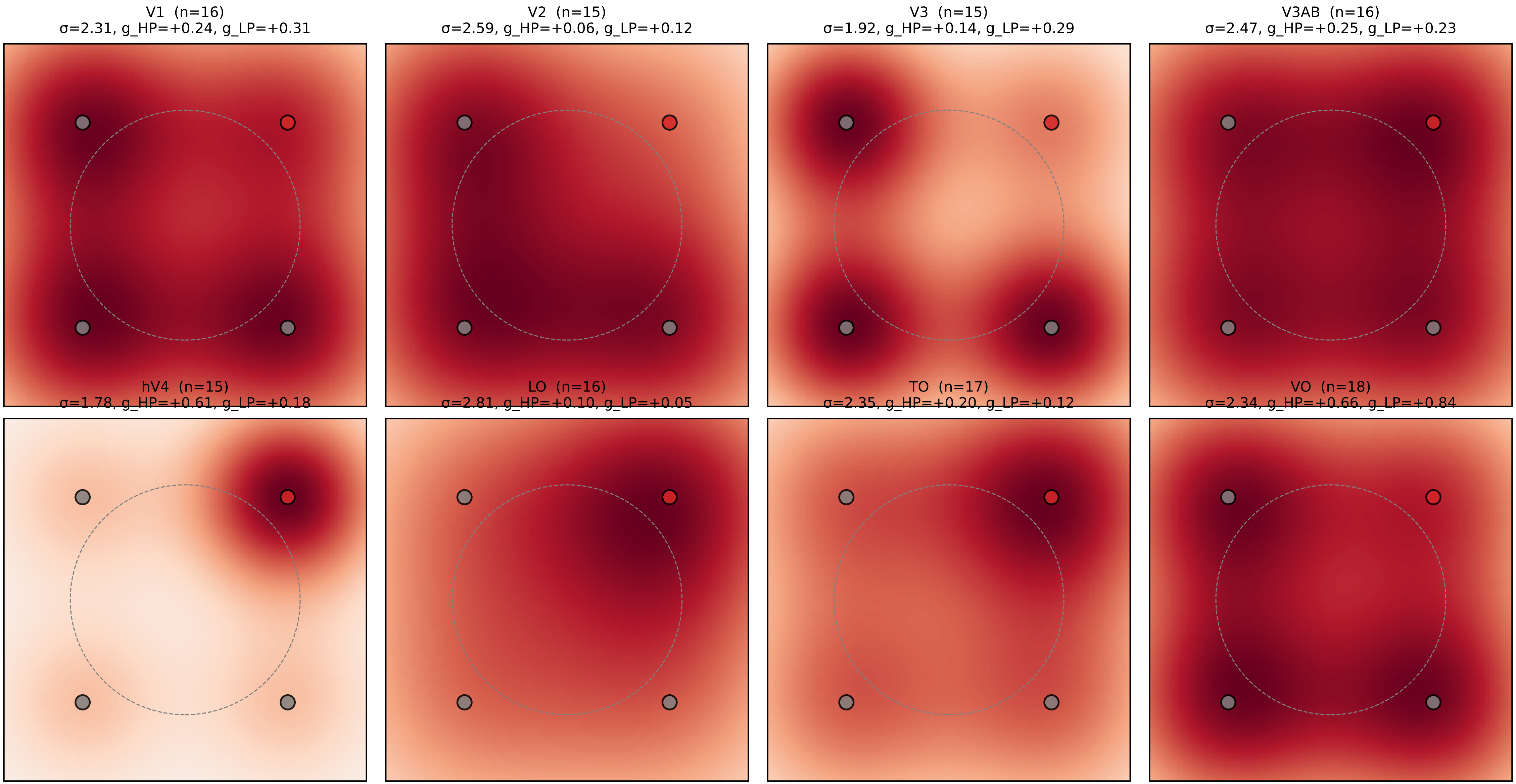


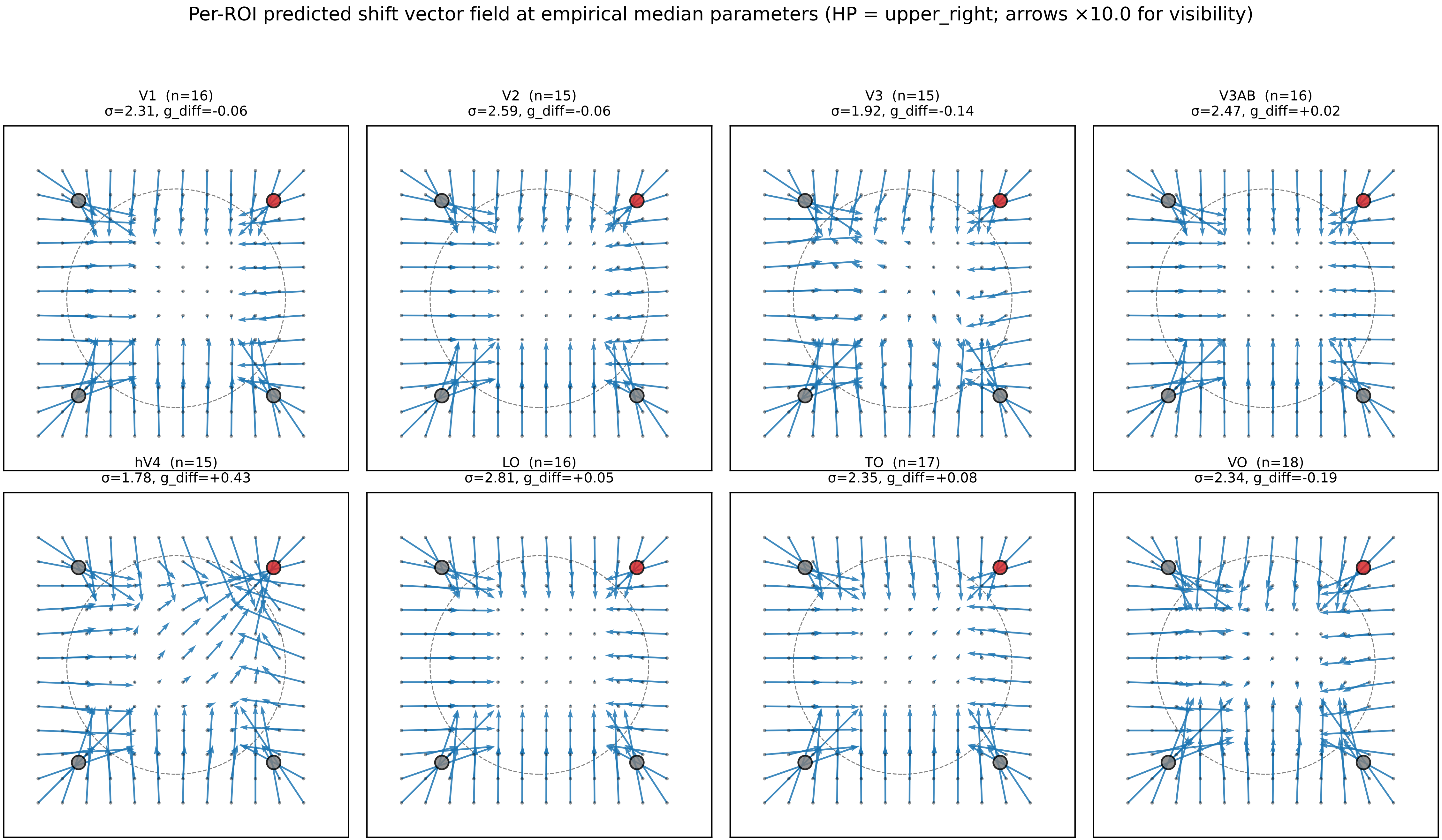
Why projection on the away-from-HP axis is small at large distance. (Aperture-only — voxels with |position| > 3.17° masked.)
σ_AF=1.92, g_HP=+0.14, g_LP=+0.29, prf_sd=1.0. HP = upper-right (red star).

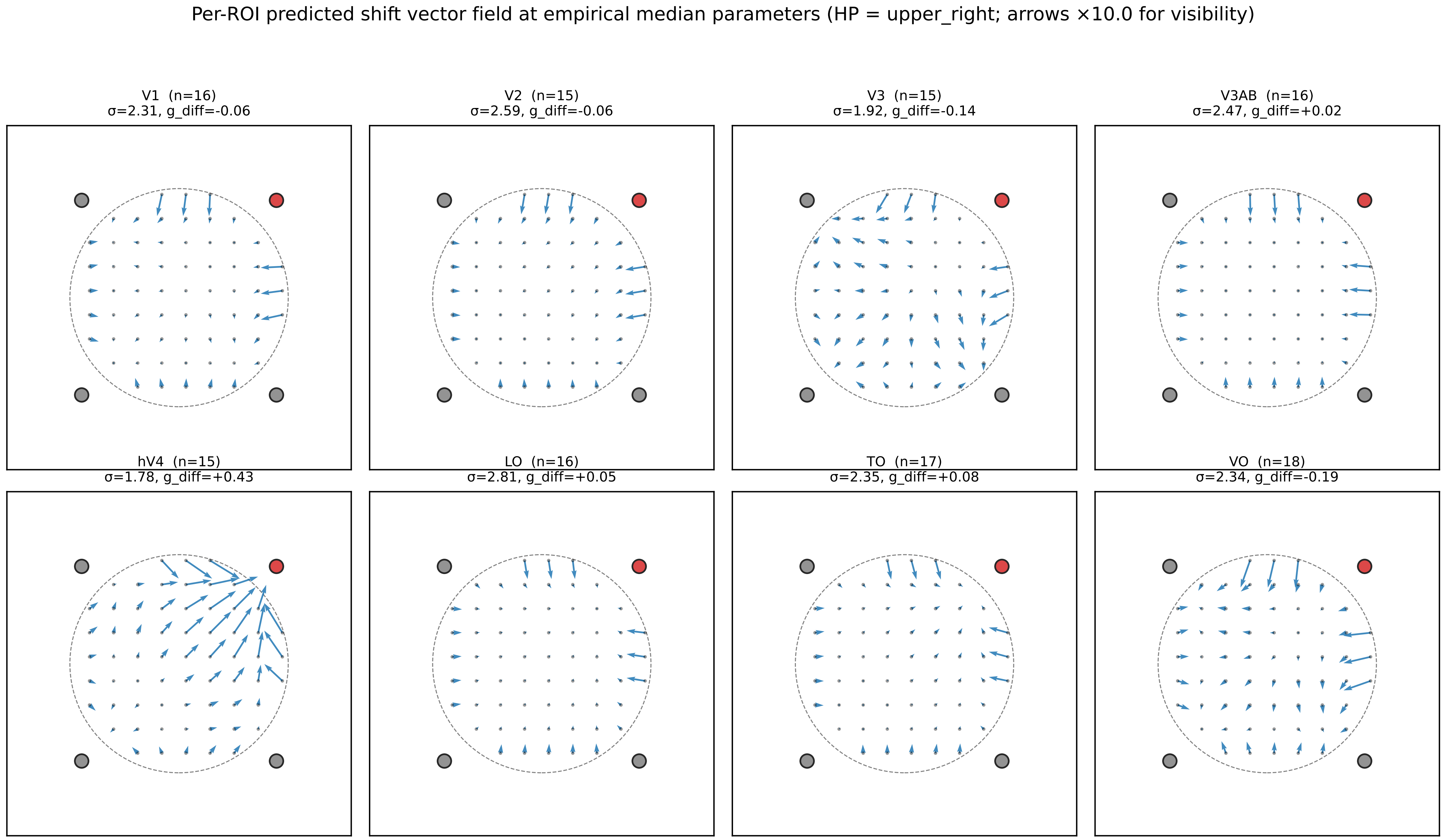



Predicted shift along the HP-axis line (HP at d=0, opposite LP at d≈5.7°).
Total |shift| can be non-zero at far distance, but its projection on
the HP-axis stays small — shifts there are radial wrt the LL-AF,
NOT wrt HP, so they cancel along this projection.
shift size (deg)
0.20
0.15
0.10
0.05
0.00
−0.05
0
2
4
6
8
distance along HP→opposite axis (deg, from HP)
|shift| (any direction)
projection on away-from-HP axis
HP location
opposite LP (LL)

Per-ROI modulation field at the empirical median parameters (across all subjects).
HP shown at upper-right (rotation symmetry — only sign of g_HP − g_LP matters).




Per-ROI predicted shift vector field at empirical median parameters (HP = upper_right; arrows ×10.0 for visibility)
V1 (n=16)
σ=2.31, g_diff=-0.06
V2 (n=15)
σ=2.59, g_diff=-0.06
V3 (n=15)
σ=1.92, g_diff=-0.14
V3AB (n=16)
σ=2.47, g_diff=+0.02
hV4 (n=15)
σ=1.78, g_diff=+0.43
LO (n=16)
σ=2.81, g_diff=+0.05
TO (n=17)
σ=2.35, g_diff=+0.08
VO (n=18)
σ=2.34, g_diff=-0.19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TO (n=17)
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VO (n=18)
σ=2.34, g_diff=-0.19